Experiments 1, 2, and 3 Processed Data

This folder contains processed data outputted from running the code.

However, if you wish to plot the data without running the code, you can download this folder from the FigShare link and follow the directory tree below to run the plotting code on your local machine.

```
$ Directory tree
.
├── outputs\
    ├── acceleration_sd
    │   ├── acc_dist_s4_cmj.pkl
    │   |── acc_dist_s4_cmj_hh.pkl
    │   |── ...
    │   └── acc_dist_s25_walking_ll.pkl
    ├── exercise_index
    │   ├── s4_exercise_index.xlsx
    │   |── s5_exercise_index.xlsx
    │   |── ...
    │   └── s25_exercise_index.xlsx
    ├── joint_angles
    │   ├── mocap
    │   │   ├── ik_s4_cmj.pkl
    │   │   ├── ik_s4_drop_jump.pkl
    │   │   ├── ...
    │   │   └── ik_s25_walking_x.pkl
    │   |── mt
    │   │   ├── ik_s4_ekf_6D_cmj.pkl
    │   │   ├── ik_s4_ekf_6D_drop_jump.pkl
    │   │   ├── ...
    │   │   └── ik_s25_xsens_9D_walking_x.pkl
    │   |── mvn
    │   │   ├── ik_s4_xsens_9D_running_x.pkl
    │   │   ├── ik_s4_xsens_9D_sts_x.pkl
    │   │   ├── ...
    │   │   └── ik_s25_xsens_9D_walking_x.pkl
    │   └── opensense
    │       ├── ik_s4l_vqf_6D_long_walk1.pkl
    │       ├── ik_s4l_xsens_9D_long_walk1.pkl
    │       ├── ...
    │       └── ik_s25_xsens_9D_walking_x.pkl
    ├── magnetic_sd
    │   ├── mag_dist_s4_cmj.pkl
    │   |── mag_dist_s4_drop_jump.pkl
    │   |── ...
    │   └── mag_dist_s25_walking.pkl
    ├── new_long_walk_opensense
    │   ├── ik_s4l_vqf_6D_long_walk1.pkl
    │   |── ik_s4l_xsens_9D_long_walk1.pkl
    │   |── ...
```

```
    │   └── ik_s23l_xsens_9D_long_walk1.pkl
    ├── os_ik
    │   ├── s4_VQF_orientation.sto
    │   |── ik_s4_VQF_orientation.mot
    │   |── ...
    │   └── s13l_Xsens_orientation.sto
    ├── rmse
    │   ├── eval_s4_ekf_6D_cmj_direct_alignment_mt.pkl
    │   |── eval_s4_ekf_6D_drop_jump_direct_alignment_mt.pkl
    │   |── ...
    │   └── eval_s25_xsens_9D_walking_x_direct_mvn.pkl
    ├── rmse_long_walk
    │   ├── s4l_mad_9D_long_walk1_mt.pkl
    │   |── s4l_mad_9D_long_walk1_mt_chunk.pkl
    │   |── ...
    │   └── s23l_xsens_9D_long_walk3_os_chunk.pkl
    ├── rmse_long_walk_ik
    │   ├── s5l_long_walk1_mc_ik.mot
    │   |── s5l_long_walk2_mc_ik.mot
    │   |── ...
    │   └── s5l_xsens_9D_long_walk3_mt_ik.mot
    ├── run_time
    │   └── mt
    │       ├── ik_s4_ekf_6D_cmj.pkl
    │       ├── ik_s4_ekf_6D_drop_jump.pkl
    │       ├── ...
    │       └── ik_s25_xsens_9D_walking.pkl
    └── sync_info
        ├── sync_info_s2_long_walk.pkl
        |── sync_info_s3_long_walk.pkl
        |── ...
        └── sync_info_s25_walking_x.pkl
```

In this folder,

- `acceleration_sd` contains variance of accelerometry data.
- `exercise_index` contains segmentation indices of each exercise for all participants. Of note, this folder is needed for evaluation codes.
- `joint_angles` contains kinematics obtained from different sources, including `mocap` (marker-based), `mt` (IMU), `mvn` (IMU with the MVN biomechanical model), and `opensense` (IMU with the OpenSense biomechanical model).
- `magnetic_sd` contains variance of magnetometer data.
- `new_long_walk_opensense` contains IMU kinematics derived from using the OpenSense biomechanical model for long-duration trials.
- `os_ik` IMU kinematics outputted from the OpenSense software before being stored in `joint_angles/opensense`.
- `rmse` contains RMSDs calculated from various comparisons.
- `rmse_long_walk` contains RMSDs calculated from various comparisons during the long-duration trials.
- `run_time` contains execution time recorded when running different state-estimation filters.

- `sync_info` stores syncing indices of IMU and marker-based data.